

Trade Restructuring

Assessing Labor Market and Welfare Effects

Kaleb Abreha

Gladys Lopez Acevedo



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Abstract

This paper assesses patterns and drivers of current trade restructuring and its welfare implications. The main trade restructuring drivers include lower cost advanced technologies, rising offshore labor costs, and recent shocks like COVID-19, trade disputes, and geopolitical tensions. Data on bilateral trade flows show that the United States and the European Union have reoriented their trade relationships. Between 2017 and 2023, for example, U.S. imports from countries like Mexico and Viet Nam grew significantly, whereas imports from China and Japan declined significantly. Market reallocation stems from tariffs, trade restrictions, and large-scale industrial policies. Countries with greater competitiveness, high logistics capabilities, and technological readiness are emerging as new production

hubs. Additionally, restructuring is having significant welfare effects. Automation has increased reshoring and increased wage inequality between high- and low-skilled workers in offshoring countries, reduced export demand, and led to job and income losses in offshore countries. Furthermore, protectionist measures have, predictably, decreased welfare. U.S.-China trade tensions, for example, raised U.S. consumer costs, reduced product variety, generated small tariff revenue, and forced exporters to absorb most of the retaliatory tariffs. Looking ahead, more evidence is needed on the long-run effects from restructuring and its effects on welfare. Meanwhile, policy dialogue should focus on preventing trade fragmentation and mitigating adverse welfare effects.

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Trade Restructuring: Assessing Labor Market and Welfare Effects*

Kaleb Abreha (Texas A&M)

Gladys Lopez Acevedo (World Bank)

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1. Introduction

Global production, manufacturing in particular, has become more interconnected and fragmented where enterprises organize production processes and business operations across spatial and functional dimensions along global value chains (GVCs). These decisions involve deciding what tasks to keep in house and which to outsource, including location choices. Between the 1990s and the 2008 “Great Recession”, substantial growth of offshoring globally involved relocating low-skilled production to developing countries (Baldwin 2016). This rapid growth of offshoring has provided countries with opportunities to reduce production costs, boost productivity and upgrade industries, while creating jobs, reducing poverty, and promoting more inclusive economic growth.

However, this trend is reversing. The 2008 Great Recession slowed offshoring, triggering “reshoring” as advanced economies sought to bring back certain manufacturing and activities. One of the main driving forces behind reshoring—and, more generally, trade restructuring—has been the increasing availability and affordability of advanced production technologies, making reshoring more feasible. Additionally, offshoring is no longer delivering the same level of cost savings—labor costs in China have increased, for example.

Simultaneously, in response to rising income inequality, job displacement, and job polarization triggered by offshoring,¹ the political economy of the main offshoring economies has leaned toward protectionism to bring back manufacturing jobs and keep R&D in-house or in “safer” countries to safeguard innovations. In addition, the COVID-19 pandemic significantly disrupted supply chains, and building more resilient production and trading systems has become a top priority in national security and development policies of most countries. Trade disputes, energy cost hikes, and global shipping challenges due to ongoing wars in Ukraine and the Middle East have further pushed countries to reassess trade policies and priorities to enhance domestic production capacity and reduce supply chain risks.

Although there have been extensive studies on offshoring, comprehensive studies on recent trade policies and how they affect welfare are just emerging. This paper assesses the patterns and drivers of current trade restructuring and their welfare implications, drawing on findings related to reshoring and protectionist measures. In so doing, it provides evidence of likely effects, especially in the short run, and identifies focus areas for policy dialogue and areas for future research.

¹ Offshoring increased wage inequality and job displacement, leading to unfavorable sentiment toward trade (Pavcnik 2017).

This work is important for at least two reasons. First, reshoring—and trade restructuring, more generally—is expected to increase over time due to technological advancements and socio-political developments in the main offshoring countries. Second, trade restructuring entails relocation of production from offshore countries or trade partners to new host countries. This could promote job creation and income gains in the new host countries but may cause production and job losses in offshore countries or trade partners.²

In addressing the main question, the paper proceeds in three steps:

First, we assess the extent to which the United States (U.S.) and European Union (EU) markets are undergoing trade restructuring. The paper shows that both are undergoing major restructuring.³ Between 2017 and 2023, U.S. imports from China declined substantially, while Mexico, Viet Nam, and Bangladesh increased their market shares. In the EU market, the Russian Federation experienced significant trade contraction, whereas the Republic of Korea, India, Brazil, Norway, and China saw growth. We also estimate a standard gravity model to identify how bilateral trade flows change based on given changes in trade policy. The estimation results show that a 1 percent increase in tariffs results in approximately a 7.25 percent decrease in total trade in the case for the U.S. and a 4.67 percent decrease for the EU.

Second, we identify the characteristics of countries that stand to gain from trade restructuring. The paper finds three key factors that influence whether countries gain or lose market shares: (a) labor productivity or unit labor cost; (b) logistics capability, including quality of trade and transport infrastructure, logistics services, and the efficiency of clearance processes; and (c) technological readiness, encompassing the availability of credit and skills, ICT infrastructure, capabilities in high-tech manufacturing and digital service delivery, and R&D. The evidence suggests that countries with higher competitiveness, enhanced logistics capability—especially in trade and transport infrastructure—and greater technological readiness (notably access to finance and industrial capacity) are better positioned to gain market shares and become new production hubs.

Finally, trade restructuring is shown to have significant welfare effects. Productivity gains from automation have led to increased reshoring, but without creating jobs or raising wages for low-skilled

² Attracting and retaining offshoring-related jobs—typically allocated to locations with the lowest costs—might lead countries to take measures at the expense of labor (e.g., lower wages, unstable contracts, and poor working conditions).

³ It is worth noting that this trade restructuring might not necessarily lead to decoupling, at least in the short-run, as countries remain integrated through GVCs. For instance, textiles and semiconductors are intermediate goods, which are central to the production of apparel and wide-ranging knowledge-intensive products, respectively.

workers while increasing wages for high-skilled workers. Automation in the U.S., for example, reduced demand for exports from countries like Colombia and Mexico, resulting in employment and earnings losses. Sectors with high levels of automation in the U.S. and local labor markets with strong export relationships were particularly affected. Domestic automation has also led to wage declines. Furthermore, it is found that protectionist measures have caused welfare losses. For example, the U.S.-China trade war increased consumer costs, reduced product variety, generated minimal tariff revenue, and forced exporters to absorb most of the retaliatory tariffs. Importantly, evidence shows the adverse effects are more severe for less-skilled, manufacturing, female, and those in smaller enterprises, and are unevenly distributed across locations within countries.

The rest of the paper is organized as follows. Section 2 discusses the main data sources and empirical approach. Section 3 discusses the patterns and drivers of trade and summarizes the evidence on labor market and welfare effects. Section 4 concludes.

2. Data and Approach

Data

The main data come from the United Nations Comtrade and contain records of product-level bilateral trade flows. This is combined with the Centre d'Études Prospectives et d'Informations Internationales (CEPII) dataset on geographic distance, sociocultural proximity, trade facilitation (GATT/WTO membership, regional trade agreements), and country characteristics. Data on the most-favored nation (MFN) and applied tariffs are collected from the World Bank's World Integrated Trade Solution (WITS) database. Additional datasets used include the World Bank's data on countries' logistics capabilities, the International Labour Organization's (ILO's) labor force surveys for data on labor productivity, and the United Nations Conference on Trade and Development's (UNCTAD's) data on frontier technology readiness.

Approach

The analysis proceeds in three steps. The first step is to characterize trade patterns in the main trading economies, namely the EU and the United States, between 2017 and 2023. We select the year 2017 as a reference since it was right before the onset of significant U.S.-China trade tensions. Additionally, to capture trade restructuring heterogeneity, we focus on textiles, apparel, and

semiconductors, sectors with high production fragmentation and global supply chain integration that are critical for economic development, productive capability, and national security.

To examine how bilateral trade flows respond to trade costs (such as geographic and cultural proximity), trade agreements, and tariffs, we estimate a standard gravity model:

$$trade_{i,j,t} = \beta_1 \log(1 + tariff)_{i,j} + \tau_{i,t} + \delta_{j,t} + \mu_{i,j} + \epsilon_{i,j,t} \quad (1)$$

where $trade_{i,j,t}$ is trade between country i and j in year t , and $tariff_{i,j,t}$ is an applied tariff.

Following the predictions of a structural gravity model, equation (1) includes country-pair fixed effects (μ_{ij}), exporter-year fixed effects ($\tau_{i,t}$), and importer-year fixed effects ($\delta_{j,t}$) to control for time-varying and time-invariant factors affecting trade flows. Equation (1) is estimated using a Poisson pseudo maximum likelihood (PPML) estimator, which addresses possible key estimation issues from trade data, such as high prevalence of zero trade flows and heteroskedasticity (Silva and Tenreyro 2006; France and Zylkin 2024).

The next step explores the main factors that likely determine which countries gain or lose market shares from trade restructuring. This involves summarizing the characteristics of countries that experienced marked changes in exports to the EU and U.S. markets. Key characteristics include indicators of competitiveness, trade infrastructure, and technological capability.

The final step draws on the empirical evidence and summarizes the labor market and welfare effects. We consider repercussions stemming from technology-induced reshoring and recent protectionist measures. Given heterogeneity across countries, we present key implications for developed and developing countries when necessary. For instance, for developed countries, job creation, wage inequality, consumer prices, and product variety are key indicators of welfare effects; while for developing countries, export growth and the number and quality (e.g., wages, working conditions, formality of employment) of jobs created are more important.

3. Results

3.1 Trade Restructuring

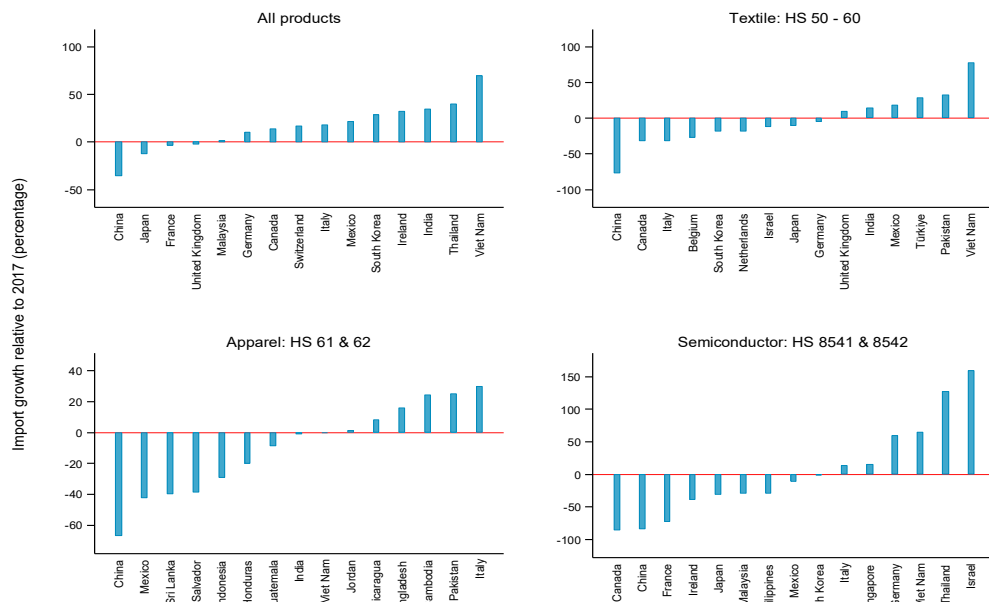
Patterns of Trade Reorientation

In recent years, the U.S. and EU exhibited significant trade reorientation, with a marked shift in sources of imports toward emerging economies in North America and Asia (other than China). Figure 1 shows that U.S. imports from Viet Nam, Thailand, India, Ireland, Korea, and Mexico grew substantially from 2017 to 2023. Asian countries like Viet Nam, Thailand, India, and Mexico and Canada emerged as key trading partners of North America and gained market shares. In contrast, traditional trade partners like China and Japan experienced a significant decline, while others, such as the United Kingdom (U.K.) and France, declined to a lesser extent.

In the textile trade, there was a marked increase in imports from Viet Nam, Pakistan, Türkiye, and Mexico, contrasting with declines in imports from China, Canada, and other advanced economies. As for apparel, there was a notable rise in imports from Italy, Pakistan, Cambodia, Bangladesh, and Nicaragua. However, imports from China, Mexico, Sri Lanka, El Salvador, Indonesia, and Honduras saw significant reductions. In the semiconductor trade, imports from Israel, Thailand, Viet Nam, and Germany grew, while decreasing from Canada, China, France, Ireland, and Japan.

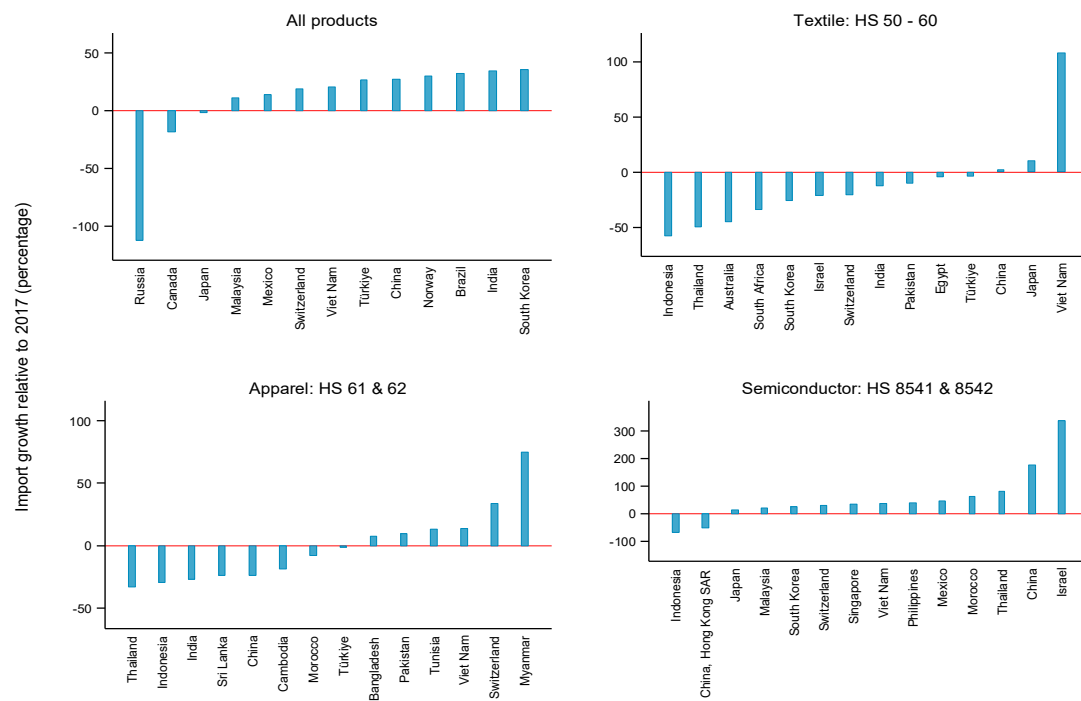
For EU imports, Figure 2 shows that Russia experienced the most significant contraction in trade, a result of geopolitical tensions. Meanwhile, imports from Korea, India, Brazil, Norway, and China grew. In the textile sector, Viet Nam outperformed other countries with significantly high growth, while China and Japan achieved relatively modest growth. However, EU textile trade with countries like Indonesia, Thailand, Australia, and South Africa decreased. On the other hand, EU apparel imports grew considerably from Myanmar (despite facing sanctions), followed by Switzerland, Viet Nam, and Tunisia, while imports from Thailand, Indonesia, India, Sri Lanka, China, and Cambodia decreased. As for semiconductors, EU imports increased from Israel, Thailand, Morocco, and Mexico, while declining from Indonesia and Hong Kong SAR, China. Unlike the U.S., semiconductor imports from China to the EU increased.

Figure 1 U.S. import growth by main trading partners and product, 2017-2023



Note: Calculations based on data from UN Comtrade. Note: Nominal values deflated using GDP deflator (2010 = 100).

Figure 2 EU import growth by main trading partners and product, 2017-2023



Note: Calculations based on data from UN Comtrade. Note: Import growth for the EU is computed based on nominal values.

Drivers of Trade Reorientation

Research has started exploring the implications of the US-China trade war that roughly started in 2018 on trade patterns and other economic outcomes. Fajgelbaum et al. (2024) provide evidence of trade reallocations. Country-specific characteristics account for about 76 percent of the cross-country growth in exports of targeted products. Analyzing which countries capture market share from China in the U.S., Freund et al. (2023) show that large, low-income countries politically aligned with the U.S., and with prior supply chain linkages to China, are more likely to gain market shares.

A large body of research has identified several key factors determining trade flows. While evidence indicates that tariff and non-tariff trade barriers and bilateral and multilateral distances reduce trade, but socioeconomic proximity such as historical ties and common language promote trade (for example, Baier et al. 2019; Abreha and Robertson 2023). Evidence also shows that trade facilitation measures, such as regional trade agreements (RTAs) promote trade, although the extent varies considerably across agreements, products, and countries involved.

Given U.S. trade tensions with China, alongside its large-scale industrial policies the CHIPS and Science Act and Inflation Reduction Act (IRA), for example, this discussion focuses on U.S. imports. We assess potential factors driving trade reorientation and what countries are likely to gain or lose market shares.⁴ It also focuses on tariff changes and factors that likely determine what countries gain or lose market shares due to policy-induced trade restructuring. We consider three factors: (i) labor productivity (a proxy for competitiveness), (ii) logistics capability (an indicator of trade and transport infrastructure quality), and (iii) technological readiness (a measure of capacity in high-tech manufacturing and digital service delivery).

Tariffs

We estimate a gravity model using bilateral trade data for all countries to assess the effect of tariffs on aggregate trade, as well as on specific products such as textiles, apparel, and semiconductors.⁵ The PPML estimates in Table 1 show that tariffs significantly reduce bilateral trade. Higher tariffs are

⁴ The annex provides the corresponding results for the EU countries.

⁵ It is worth noting that tariffs are levied on a highly disaggregated level of product classification and vary substantially even within narrowly defined product groups. Hence, the tariffs at the level of product groups are (simple) averages of the tariffs at the disaggregated level and could hide import difference for analysis at aggregate levels.

associated with reduced trade in total, textile, and apparel categories. Specifically, a 1 percent increase in tariffs corresponds to a decrease in total trade, textile trade, and apparel trade by approximately 1.23 percent, 1.87 percent, and 0.48 percent, respectively. Note also that tariff effects on textile trade is larger than for apparel. By contrast, tariffs do not seem to significantly decrease trade in semiconductors, suggesting these products encounter low tariffs in most of the bilateral trade flows or even less price elastic due to their importance. Our estimates are consistent with previous findings. While Kee et al. (2008) find that a 1 percent increase in tariffs reduces imports by an average of about 0.6 to 2.1 percent across various countries and industries, Felbermayr et al. (2013) show that a 1 percent increase in tariffs could reduce bilateral trade by approximately 0.5 percent.

Table 1 Effect of Tariff on Imports by Product Groups

	(1) Total	(2) Textile	(3) Apparel	(4) Semiconductor
log (1 + Tariff)	-1.2281*** (0.2848)	-1.8696*** (0.3799)	-0.4821** (0.1907)	2.0484 (3.0517)
Constant	23.4994*** (0.0128)	19.6810*** (0.0248)	21.0612*** (0.0150)	22.3788*** (0.0069)
Importer-year FEs	Yes	Yes	Yes	Yes
Exporter-year FEs	Yes	Yes	Yes	Yes
Importer-exporter FEs	Yes	Yes	Yes	Yes
Observations	653437	409175	482844	303234
Chi-squared	18.5983	24.2160	6.3912	0.4506
p-value	0.0000	0.0000	0.0115	0.5021

Note: Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 2 Effect of Tariff on Imports by Product Groups and Main Trading Partners

	United States				European Union			
	(1) Total	(2) Textile	(3) Apparel	(4) Semiconductor	(5) Total	(6) Textile	(7) Apparel	(8) Semiconductor
log (1 + Tariff)	-7.2462*** (1.7855)	0.8756 (1.0683)	4.4294*** (0.8700)	0.0000 (.)	-4.6622*** (0.4587)	0.9982*** (0.3443)	-5.2688*** (0.3759)	-10.6944*** (1.2314)
Constant	25.3515*** (0.0408)	20.2688*** (0.0535)	21.7266*** (0.0889)	21.6639*** (0.0376)	21.9299*** (0.0240)	17.6145*** (0.0260)	19.6191*** (0.0419)	18.6170*** (0.0366)
Exporter FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5112	4920	5040	5016	129024	93768	108720	81072
Chi-squared	16.4711	0.6718	25.9245	.	103.3153	8.4041	196.4912	75.4247
p-value	0.0000	0.4124	0.0000	.	0.0000	0.0037	0.0000	0.0000

Note: Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

We estimate the gravity model separately on trade flows to the U.S. and EU markets. In this regard, Table 2 reveals the difference in tariff changes across the two markets for each of the products. In the U.S. market, a 1 percent increase in tariffs results in approximately a 7.25 percent decrease in U.S. total trade, while it leads to a 4.66 percent decrease in total trade of the EU. These tariff effects are larger compared with those found in Table 1, particularly for U.S. imports.

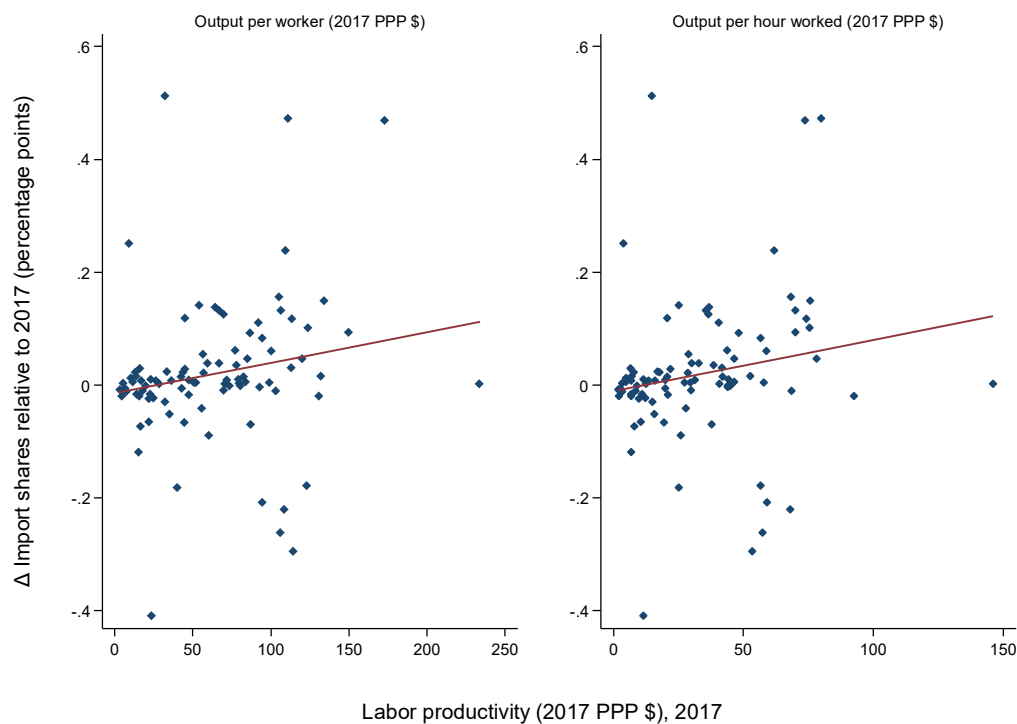
In the textile sector, tariffs do not affect U.S. imports, whereas EU textile imports have increased despite higher tariffs. For U.S. apparel trade, large statistically significant coefficient implies a 4.43 percent increase in apparel trade for a 1 percent increase in tariffs. This counterintuitive result suggests that the apparel sector has a high level of protection, and the largest exporters face high tariffs. Imports have surged despite the protective tariffs. Conversely, for the EU, a 1 percent increase in tariffs lowers apparel trade by about 5.27 percent. Tariff changes do not affect semiconductor trade on US imports, but the estimate for the EU shows that a 1 percent increase in tariffs leads to a large 10.69 percent decrease in semiconductor trade.

Overall, the gravity estimates indicate that tariffs significantly reduce trade for all countries, particularly for U.S. and EU imports. Despite differences in the magnitude of tariff effects across products and markets, the results show that recent protectionist measures such as tariffs and trade restrictions (sanctions and export controls) reduced and redirected trade.

Labor Productivity

Productivity represents a key driver of a country's economic competitiveness and export success. Labor productivity is central in determining which countries increase or decrease market shares from trade restructuring. As Figure 3 shows, countries with higher labor productivity in 2017 are more likely to have had greater market share and to have gained U.S. import share after the start of the U.S.-China trade war. In the EU market higher labor productivity is associated with greater import shares, but it is negatively correlated with changes in market share, indicating traditional partners with higher labor productivity lost market shares.

Figure 3 Change in U.S. Import shares in 2023 against labor productivity in 2017 (in thousands)



Note: Calculations based on data from the ILO, countries' productivity is measured by output per worker and output per hour worked, expressed in GDP (constant 2017 international dollars at PPP). The dots represent countries.

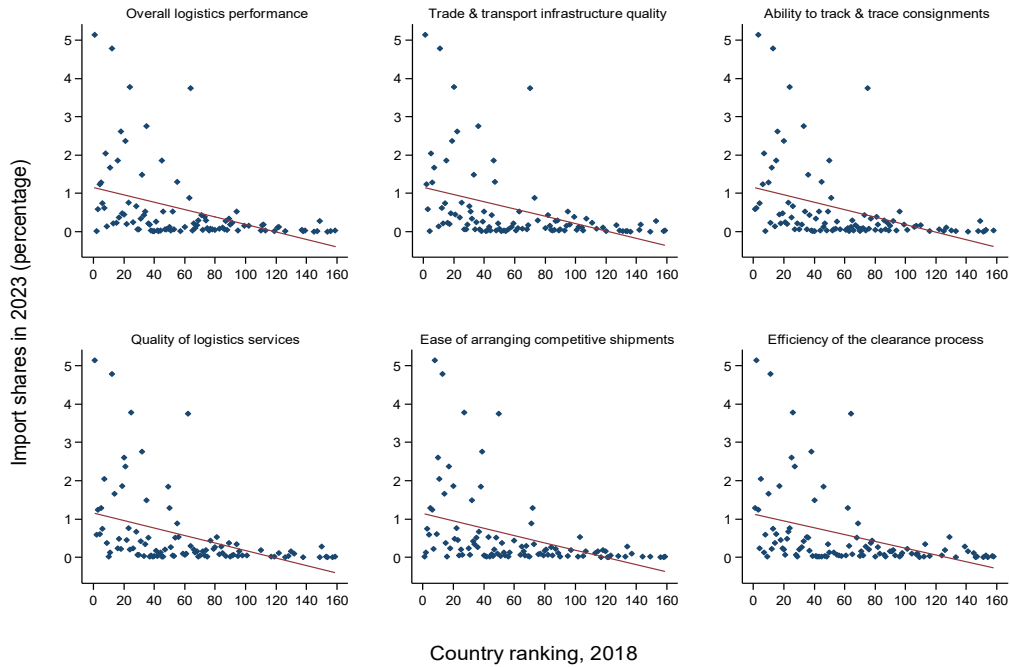
Logistics Capacity

This indicator captures a country's logistics performance based on customs clearance efficiency, trade and transport infrastructure quality, ease of arranging affordable international shipments, logistics services quality, tracking capability, and on-time delivery frequency. Figure 4 plots import market shares in 2023 against countries' rankings in logistics capacity in 2018.⁶ It illustrates that higher logistics capability is associated with a larger market share. Analysis of various components of logistic capability uncovers that the quality of trade and transport infrastructure, ability to trace and track consignments, quality of logistics services, ease of arranging competitive shipments, and efficiency of clearance processes correlate with countries' U.S. imports share. Moreover, logistics capability plays a pivotal role in terms of changes in market share; that is, countries with better logistics capability in 2018 are likely to

⁶ The year 2018 is used as a reference because of lack of data for 2017.

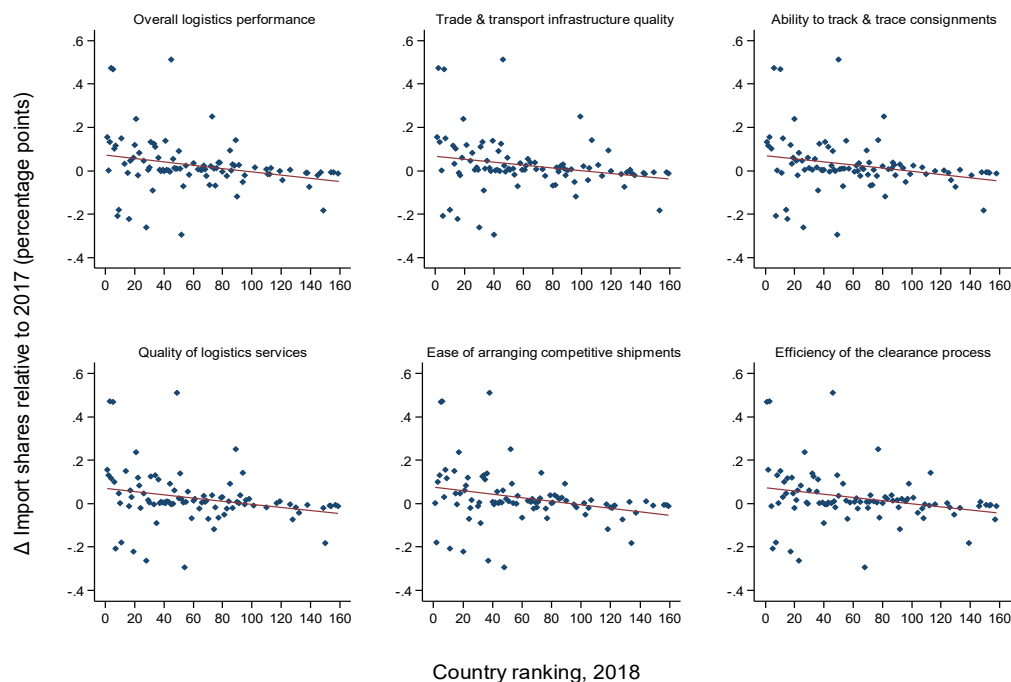
have gained market share (Figure 5). The quality of trade and transport infrastructure emerges as a primary determinant of market share gain compared to other indicators of logistics capabilities.

Figure 4 U.S. Import shares in 2023 against countries' logistics performance rankings in 2018, (1= highest performer)



Source: Calculations based on data from UN Comtrade and WDI. Note: The plot is based on the largest 100 partners in 2017. Data on the logistic performance index in 2017 is available.

Figure 5 Change in U.S. import shares during 2017-23 against countries' logistics performance ranking in 2018, (1= highest performer)



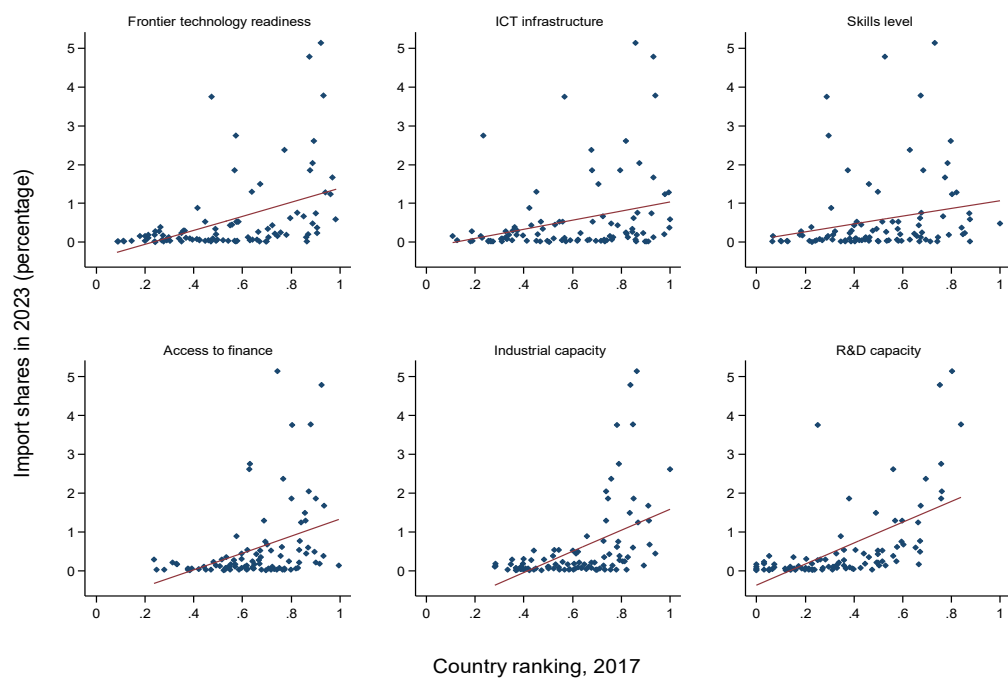
Source: Calculations based on data from UN Comtrade and WDI. Note: The plot is based on the largest 100 partners in 2017. Data on the logistic performance index in 2017 is available.

Technological Readiness

The frontier technology readiness index reflects the capacity of countries to utilize, adopt, and adapt advanced technologies. It encompasses the prevalence and quality of information and communication technology (ICT) infrastructure, availability of skills, R&D activity, capacity in key industries and industry engagement, and financial access to support technological endeavors.

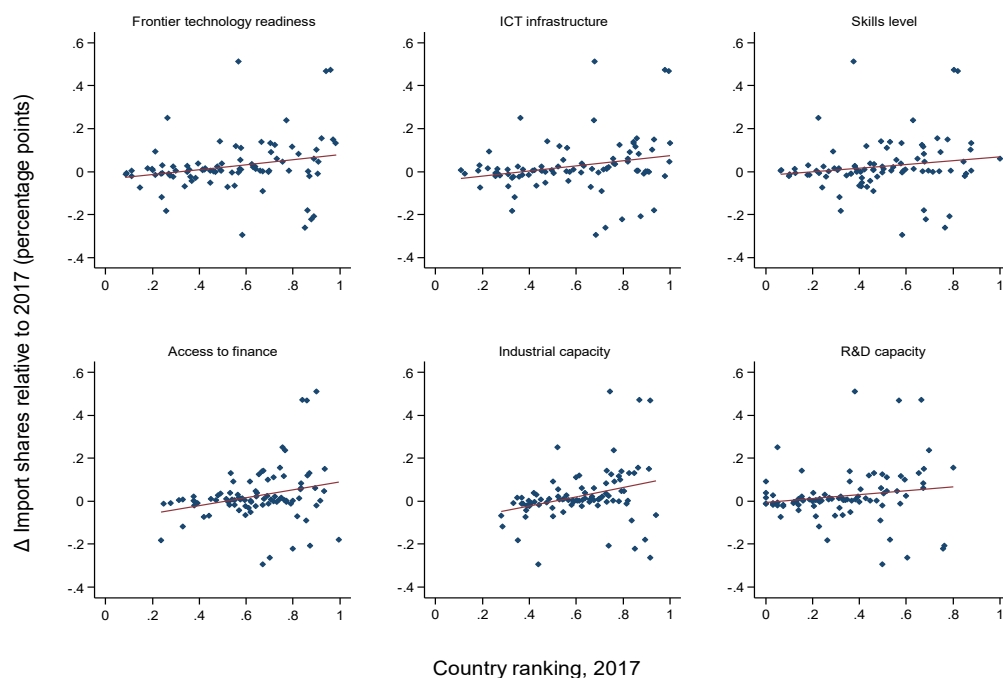
Figure 7 illustrates the relationship between U.S. import market shares and countries' rankings in 2017. Notably, higher readiness in frontier technology corresponds to higher market shares. Each of these elements correlates positively with increased market shares, and while various components of frontier technology readiness contribute to market share, access to finance and industrial capacity emerge as prominent factors.

Figure 6 U.S. import shares in 2023 against technology readiness ranking in 2017, (1= highest performer)



Note: Calculations based on data from UN Comtrade and UNCTAD. *Note:* The plot is based on the largest 100 partners in 2017.

Figure 7 Change in U.S. import shares in 2023 against technology readiness ranking in 2017, (1= highest performer)



Source: Based on data from UN Comtrade and UNCTAD. Note: The plot is based on the largest 100 partners in 2017.

To summarize, besides geography, socioeconomic proximity, and political alignment, three key factors influence countries' market gains or losses: (i) labor productivity, and more importantly unit labor costs, that is, labor cost per labor productivity; (ii) logistics capability such as trade and transport infrastructure quality, logistics service quality, and clearance process efficiency, and (iii) technological readiness in terms of readiness to effectively harness frontier technologies, including credit and skill availability, ICT infrastructure, and industrial capacity in terms of high-technology manufacturing and digital delivery of services. Hence, countries with high competitiveness, better logistics capability—especially in trade and transport infrastructure—and greater technological readiness, particularly access to finance and industrial capacity, are positioned to gain market share and become new production hubs.

The next section summarizes emerging evidence on labor market outcomes and welfare effects stemming from recent protectionist measures and technology-driven trade restructuring.

3.2 Labor Market and Welfare Implications

International trade has promoted countries' economic development by reducing poverty, creating jobs, and boosting productivity. At the same time, it has resulted in major distributional impacts associated with income inequality and job displacement. Ongoing trade restructuring involving broad reconfiguration of trade relationships, and trade and industrial policies can have serious implications, but evidence is scant of how these measures would eventually play out. To better anticipate likely welfare effects from trade restructuring, this discussion draws on findings from studies on labor market and welfare effects related to reshoring and recent trade protectionist measures.

Employment and Wage Effects

International trade affects workers differently, primarily depending on their employment sector and skill profile. Specifically, countries' GVC participation is linked with increased routine-intensive work, particularly in offshorable occupations in the industrial sector, and led to lower wages and higher wage inequality (Lewandowski et al. 2023). This relationship is more pronounced for middle-income countries (MICs) and the manufacturing sector (Szymczak and Wolszczak-Derlacz 2022; Carneiro et al. 2023).

The rise of automation in advanced economies and the resulting productivity growth have driven reshoring. An increase of one robot per 1000 workers is associated with a 3.5 percent increase in reshoring activity (Krenz et al. 2021). Such reshoring has not led to more employment or higher wages for low-skilled workers. However, reshoring created jobs for high-skilled workers and increased wages since these workers are complementary to automation within production facilities (Bolter and Robey 2020), and this has benefited high-skilled workers and increased wage inequality.⁷ Evidence also shows adverse employment effects in developing countries' labor markets, especially for low-skilled workers, women, and workers in manufacturing and smaller enterprises. Faber (2020) shows a significant negative impact of U.S. robots on employment in Mexico between 1990 and 2015, driven by the fact that production was reshored back to the U.S., displacing demand for exports from Mexico. Again, these effects are stronger for semi-skilled workers in manufacturing. These effects are also observed in other emerging economies (Pavez and Martinez-Zarzoso 2023).

⁷ This is consistent with studies that argue the availability of skilled labor is a key factor behind reshoring decisions (e.g., Pourhejazy and Ashby (2021)).

In the case of Colombia, Kugler et al. (2020) find that U.S. automation replaced exports from Colombia with cheaper robot-made U.S. products, with an estimated loss of 63,000 to 100,000 jobs between 2011 and 2016, including more dismissals, lower hires, and earning losses. This effect is stronger for workers in sectors with high levels of automation in the U.S. labor market and local labor markets with strong export relationships with the U.S. Higher robot exposure also leads to lower formal employment accompanied by a rise in informal employment in Argentina, Brazil, and Mexico, particularly among young and semi-skilled workers (Brambilla et al. 2023).

In addition, domestic deployment of industrial robots has led to wage declines in China, and less skilled workers are most affected (Giuntella et al. 2022). Likewise, Artuc et al. (2019) show that domestic automation in Mexico does not significantly affect total employment, yet it lowers the fraction of those in wage employment and increases those in the informal sector (which absorbs most displaced workers). Furthermore, automation in domestic sectors shows spillover effects and tends to negatively affect employment and wages in other sectors (Pavez and Martínez-Zarzoso 2023).

Assessing labor market effects across sectors and locations within countries is equally important. Despite the insignificant overall impact of U.S. automation on Mexican wage employment or manufacturing wage employment, Artuc et al. (2019) show that it has decreased manufacturing wage employment in locations where occupations were more exposed to automation, but increased manufacturing wage employment in other locations.

Welfare Effects

Recent trade and industrial policy measures in countries have had significant welfare implications. It is plausible that trade restructuring could help mitigate supply chain risks, which ensures product availability and stabilizes consumption (Cohen et al. 2022). On the other hand, restrictive trade measures could raise costs for consumers and cause welfare loss. For example, Flaaen et al. (2020) show that the 2018 U.S. tariffs increased the median price of washers and dryers by about US\$86 and US\$92 per unit, respectively. This resulted in an annual increase of over US\$1.5 billion in consumer costs, about US\$82 million in tariff revenue, and approximately 1,800 jobs.

Similarly, U.S.-China trade tensions are associated with significant costs to consumers. Amiti et al. (2019) estimates this cost at about US\$8.2 billion in terms of reduction in real income, and an additional

transfer of US\$14 billion in tariff revenue to the government (since prices received by foreign exporters remained largely unchanged). Besides increases in consumer prices and markups (the average price of U.S. manufacturing by 1 percentage point), the trade war reduced the variety of products available to consumers due to trade redirection of about US\$165 billion of trade annually due to import substitution from targeted to untargeted countries and products.

The 2018 U.S. import tariffs and the retaliatory tariffs on U.S. exports also reduced total trade (imports and exports). U.S. imports at the country-product level declined about 32 percent, with no significant change occurring in before-duty import prices of products coming from targeted and untargeted countries. This suggest that import substitution across source countries, and complete pass-through of the tariff burden (Fajgelbaum et al. 2019). At the same time, U.S. exports declined about 10 percent due to retaliatory measures from targeted countries (measures by China, Canada, EU, and Mexico, for example). Overall, the tariff measures resulted in a real income loss of approximately US\$51 billion, or 0.27% of GDP, for U.S. consumers and firms that rely on imported inputs. After accounting for tariff revenues and domestic producer gains, the net aggregate real income loss was about US\$7.2 billion, or 0.04% of GDP.

Relatedly, Cavallo et al. (2021) found that import tariff pass-through was much higher than exchange rate pass-through, in which Chinese exporters did not significantly lower their prices even when the dollar appreciated. However, U.S. retail prices have not changed much, implying that tariff incidence has largely fallen on U.S. firms.⁸ The longer the import tariffs are enforced, the higher the chances this will lead to lower before-duty U.S. import prices or higher pass-through into consumer prices. In contrast, U.S. exporters significantly lowered prices affected by foreign retaliatory tariffs. Waugh (2019) also estimates the effect on consumption growth (measured by new auto sales) due to Chinese retaliatory tariffs, and counties with high exposure to retaliatory tariffs (defined by industry employment shares) saw a 3.8 percentage point decline in consumption growth, attributed to the decline in tradeable and retail employment.

In general, trade restructuring and its driving factors are likely to persist, leading to significant distributional effects both across and within countries. While new host countries may experience economic development and job creation, offshore countries could face production and job losses.

⁸ Fajgelbaum et al. (2019) note the need for more evidence on the degree of tariff pass-through due to the trade war.

Whether the restructuring will achieve its intended goals remains to be seen, but recent findings suggest that protectionist trade policies are costly for consumers and firms in the short run. Despite the growing body of evidence on labor market outcomes and welfare effects, more research is needed to assess the full long-run implications of trade restructuring and accompanying policy measures, including the mechanisms and aggregate and distributional effects.

4. Conclusion

Technological advances, shifting economic policies, and evolving supply chain dynamics have markedly affected the landscape of global trade and production. Trade restructuring, driven by rising overseas labor costs, automation, and a desire for greater supply chain control, has become increasingly significant. It addresses vulnerabilities exposed by global disruptions such as the COVID-19 pandemic and geopolitical tensions, prompting firms to reconsider production strategies.

Currently, the U.S. and the EU are trade restructuring with tariffs and sanctions as driving forces along with large-scale industrial policies. Protectionism and retaliatory measures might lead to trade fragmentation, which shrinks GDP and undermines economic convergence between high- and low-income countries. Strong trade integration also remains vital for sustaining post-COVID-19 recovery by improving access to food, production inputs, and other essential goods. To address challenges stemming from global disruptions—such as the COVID-19 pandemic and geopolitical tensions—and to leverage opportunities, policy dialogue should aim to avoid trade wars, strengthen economic partnerships, and integrate social policies to mitigate job and wage losses and other negative effects on workers in both advanced and developing economies. Further research is also required to understand the micro-level effects and transmission mechanisms of these policies, especially in the long run.

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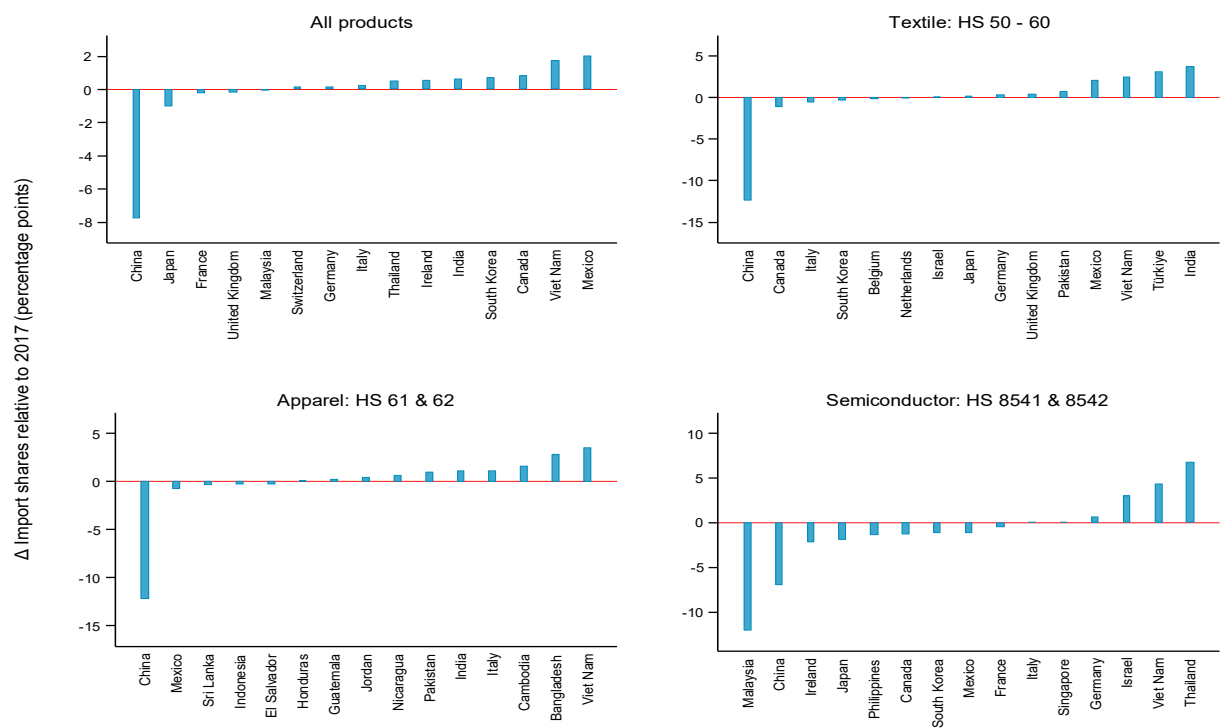
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Annex

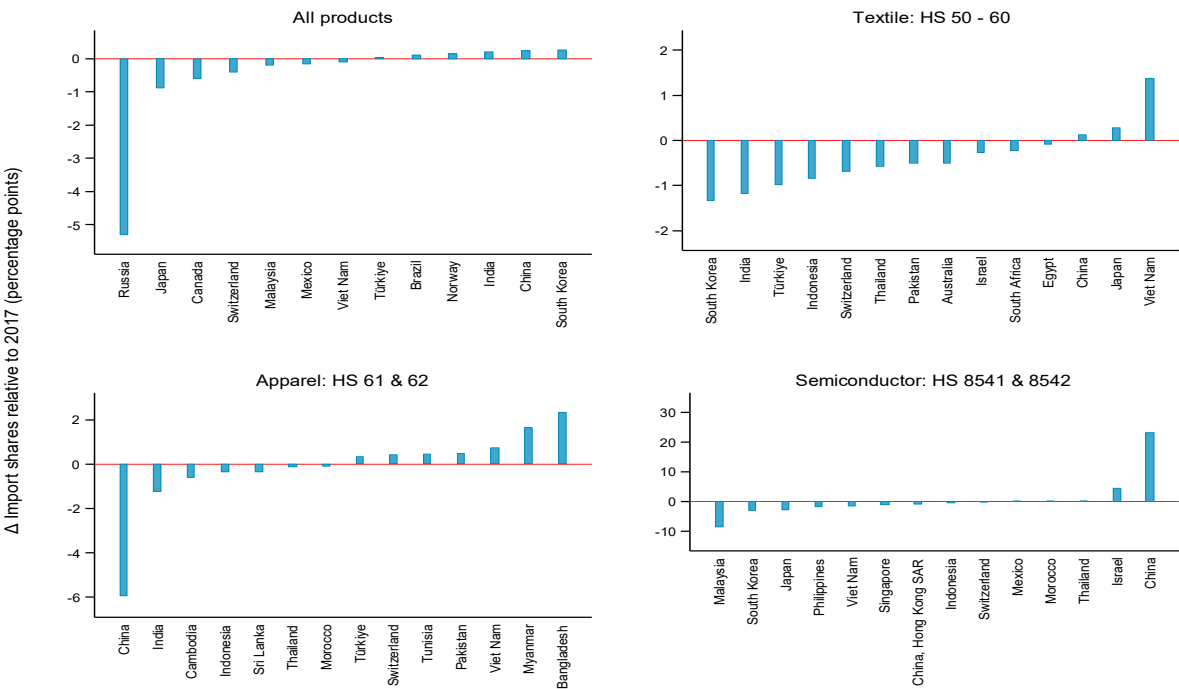
Import Shares, 2017-2023

Figure A. 1 U.S. import shares by main trading partners and product, 2017-2023



Source: Based on data from UN Comtrade.

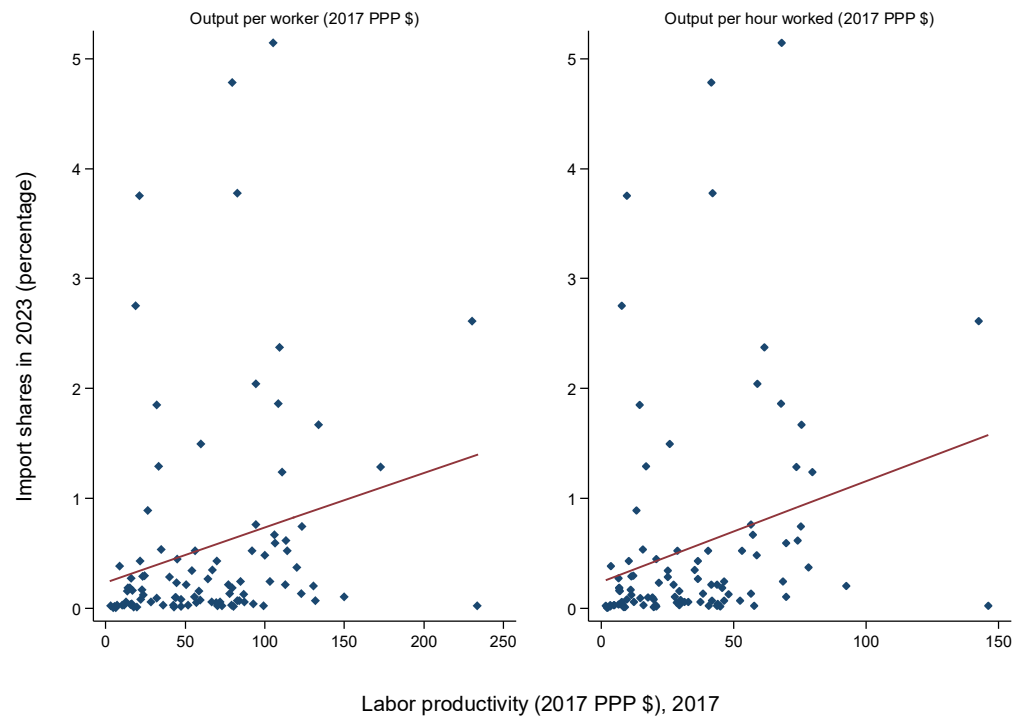
Figure A. 2 EU import shares of main trading partners by product, 2017-2023



Source: Based on data from UN Comtrade.

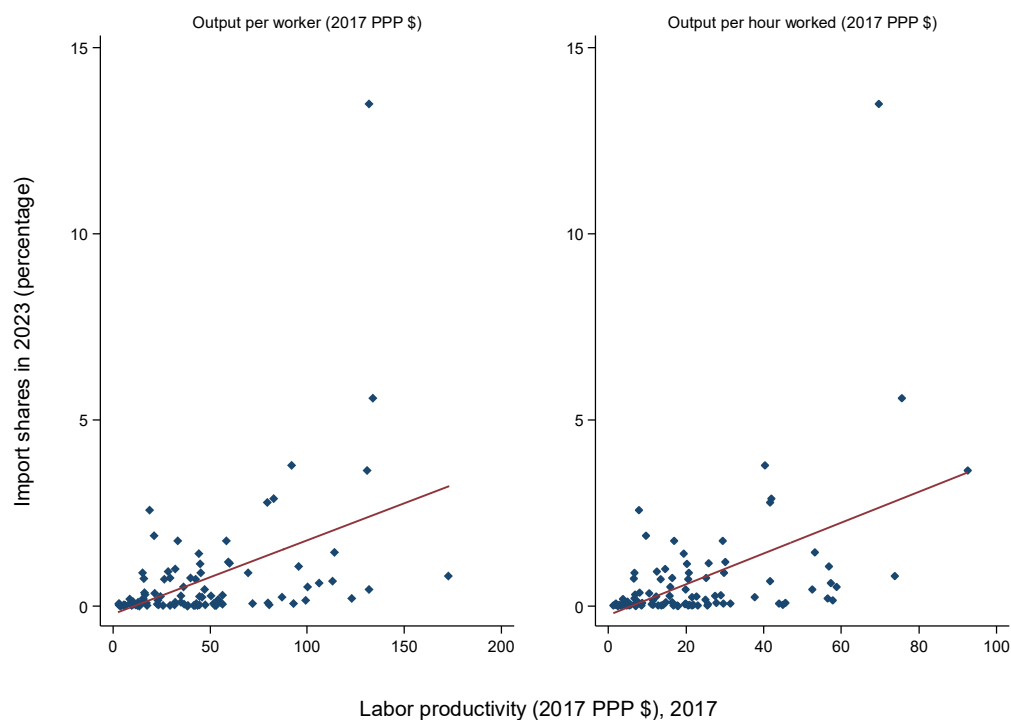
Labor Productivity

Figure A. 3 U.S. Import shares in 2023 against labor productivity in 2017 (in thousands)



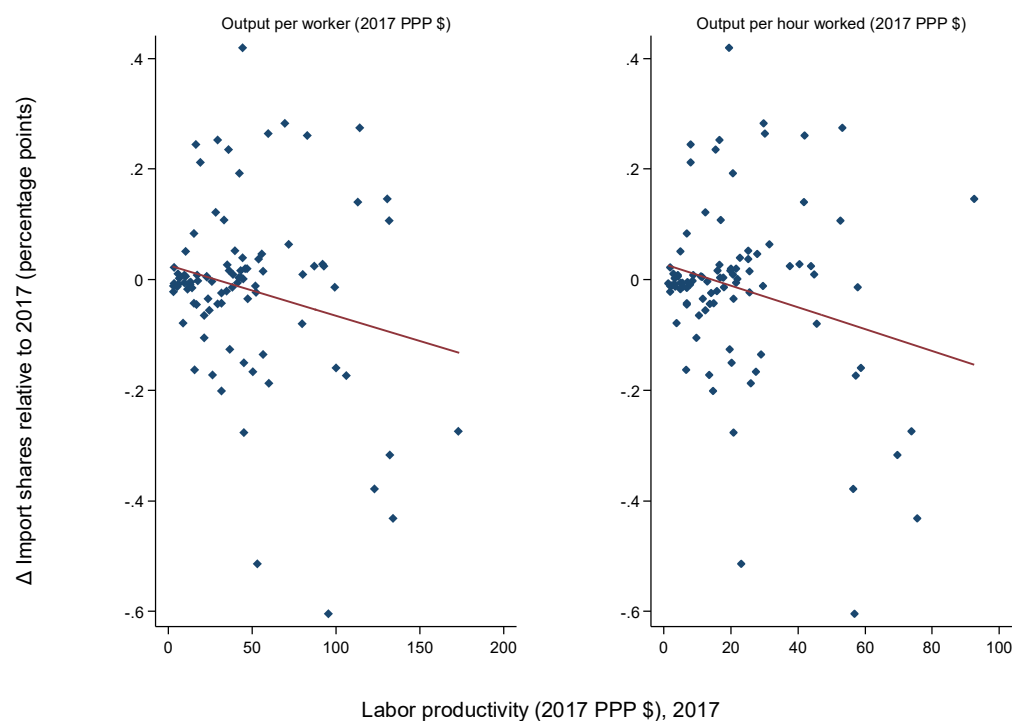
Note: Calculations based on data from the ILO, countries' productivity is measured by output per worker and output per hour worked, expressed in GDP (constant 2017 international dollars at PPP). The dots represent countries.

Figure A. 4 EU Import shares in 2023 against labor productivity in 2017 (in thousands)



Note: Calculations based on data from the ILO, countries' productivity is measured by output per worker and output per hour worked, expressed in GDP (constant 2017 international dollars at PPP). The dots represent countries.

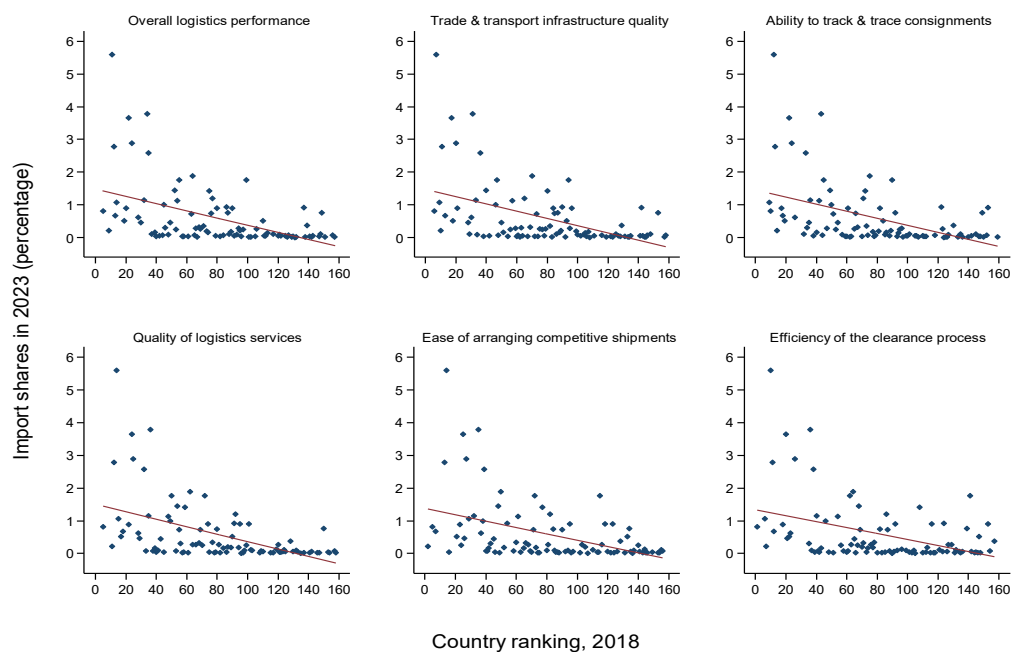
Figure A. 5 Change in EU Import shares in 2023 against countries' labor productivity in 2017 (in thousands)



Note: Calculations based on data from the ILO, countries' productivity is measured by output per worker and output per hour worked, expressed in GDP (constant 2017 international dollars at PPP). The dots represent countries.

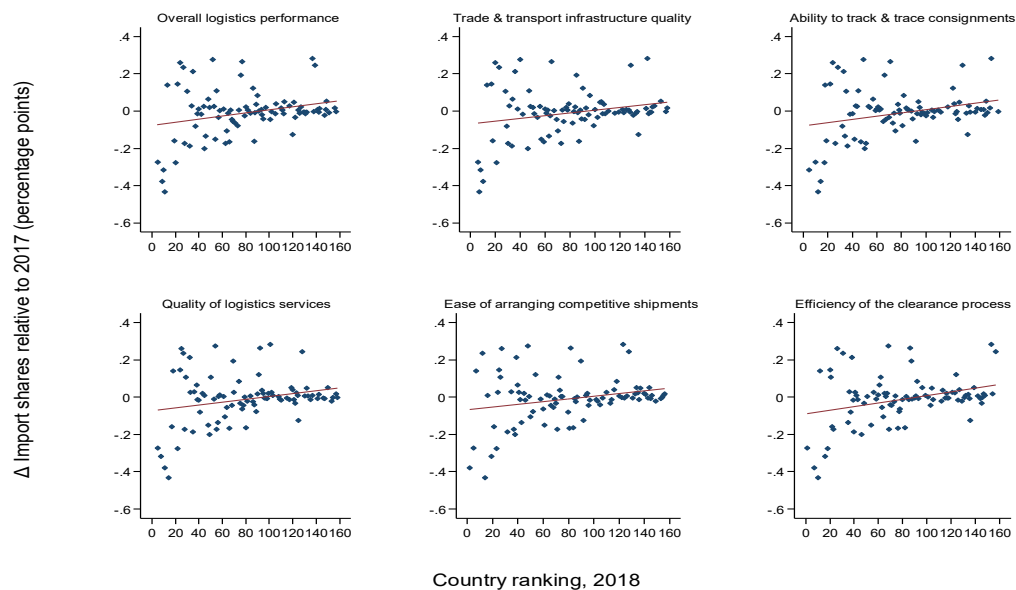
Logistics Performance

Figure A. 6 EU Import shares in 2023 against logistics performance ranking in 2018, (1= highest performer)



Note: Calculations based on data from UN Comtrade and WDI. *Note:* The plot is based on the largest 100 partners in 2017. Data on the logistic performance index in 2017 is available.

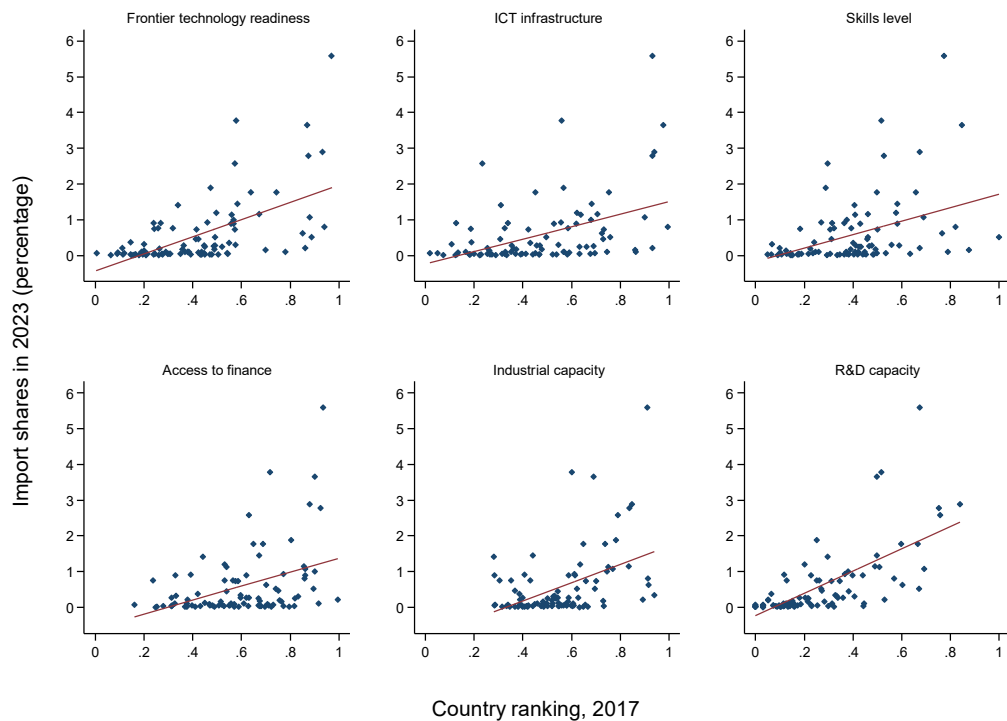
Figure A. 7 Change in EU import shares during 2017-23 against logistics performance ranking in 2018, (1= highest performer)



Note: Calculations based on data from UN Comtrade and WDI. *Note:* The plot is based on the largest 100 partners in 2017. Data on the logistic performance index in 2017 is available.

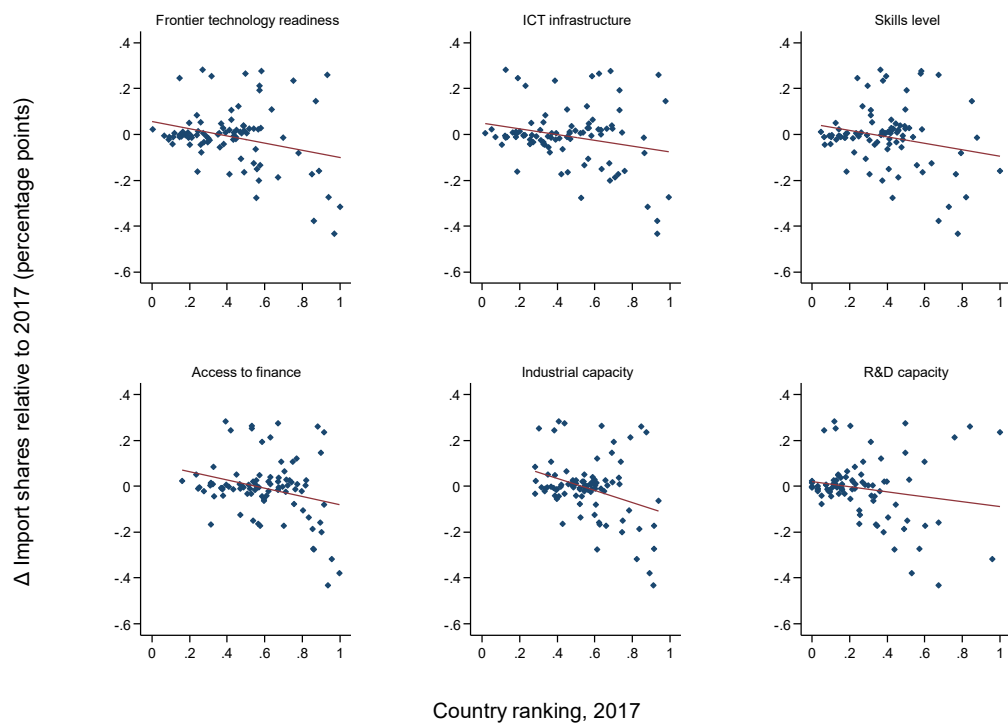
Technology Readiness

Figure A. 8 EU import shares in 2023 against technology readiness ranking in 2017, (1= highest performer)



Note: Calculations based on data from UN Comtrade and UNCTAD. Note: The plot is based on the largest 100 partners in 2017.

Figure A. 9 Change in EU import shares in 2023 against technology readiness ranking in 2017, (1= highest performer)



Note: Calculations based on data from UN Comtrade and UNCTAD. Note: The plot is based on the largest 100 partners in 2017.